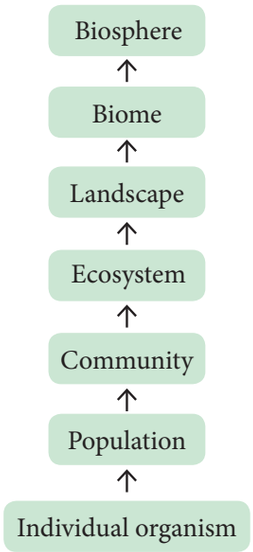
6.1.2 Ecological hierarchy

The interaction of organisms with their environment results in the establishment of grouping of organisms which is called **ecological hierarchy** or ecological levels of organization. The basic unit of ecological hierarchy is an individual organism. The hierarchy of ecological systems is illustrated below:



6.1.3 Branches of Ecology:

Ecology is mainly divided into two branches, they are autecology and synecology.

1. Autecology is the ecology of an individual species and is also called species ecology.

2. Synecology is the ecology of a population or community with one or more species and also called as community ecology.

Many advances and developments in the field ecology resulted in various new dimensions and branches. Some of the advanced fields are Molecular ecology, Eco technology, Statistical ecology and Environmental toxicology.

6.1.4 Habitat and Niche

Habitat

Habitat is a specific physical place or locality occupied by an organism or any species which has a particular combination of abiotic or environmental factors. But the environment of any community is called **Biotope**.

Niche

An ecological niche refers to an organism's place in the biotic environment and its functional role in an ecosystem. The term was coined by the naturalist **Roswell Hill Johnson** but **Grinell** (1917) was probably first to use this term. The habitat and niche of any organism is called **Ecotope**

The differences between habitat and niche are as follows.

	Habitat	Niche
1.	A specific physical space occupied by an organism (species)	A functional space occupied by an organism in the same eco-system
2.	Same habitat may be shared by many organisms (species)	A single niche is occupied by a single species
3.	Habitat specificity is exhibited by organism.	Organisms may change their niche with time and season.

Table 6.1: Difference between habitat and niche



Applied ecology or environmental technology :

Application of the Science of ecology is otherwise called as **Applied ecology or Environmental technology.** It helps us to manage and conserve natural resources, particularly ecosystems, forest and wild life conservative and management. Environmental management involves Bio-diversity conservation, Ecosystem restoration, Habitat management, Invasive species management, Protected areas management and also help us plan landscapes and environmental impact designing for the futuristic ecology.

6.1.5 Ecological equivalents

Taxonomically different species occupying similar habitats (Niches) in different geographical regions are called **Ecological equivalents.**

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